Date :

1) cabin bounce, pitch & roll reffered from the shared doc.

→ z_c

→ θ_c

→ ϕ_c

3. Chassis bounce equation → DOF-2 $\quad 2 \to r_1 (m_1)$, $3 \to r_2$

$$M_2 \ddot{z}_c + k_f (z_c - a\theta - z_{1f}) + c_f (\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})$$

$$+ k_2 (z_c - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2 \theta)$$

$$+ c_2 (\dot{z}_c - \dot{z}_2 + l_2 \dot{\theta})$$

$$+ k_3 (z_c - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DR3} + l_3 \theta)$$

$$+ c_3 (\dot{z}_c - \dot{z}_3 + l_3 \dot{\theta}) = 0.$$

4. Chassis Pitch equation → DOF 4.

$$+ I_{yx} \ddot{\phi}_s$$

$$I_{yy} \ddot{\theta} - a[k_f (z_c - a\theta - z_{1f}) + c_f (\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})]$$

$$+ l_2 [k_2 (z_c - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DL2} + l_2 \theta)$$

$$+ c_2 (\dot{z}_c - \dot{z}_2 + l_2 \dot{\theta})]$$

$$+ l_3 [k_3 (z_c - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DL3} + l_3 \theta)$$

$$+ c_3 (\dot{z}_c - \dot{z}_3 + l_3 \dot{\theta})]$$

2. Sprung mass / chassis roll :-

$$-\left(I_{xx}\ddot{\phi}_s + m_2 h_1^2 \ddot{\phi}_s\right) + I_{xy}\ddot{\theta}$$

$$= -m_2 g h_1 \phi_s + k_{rf}(\phi_s - \phi_f)$$

$$+ c_{rf}(\dot{\phi}_s - \dot{\phi}_f)$$

$$+ k_{r1}(\phi_s - \phi_{r1} - \phi_{NRS}) + c_{r1}(\dot{\phi}_s - \dot{\phi}_{r1})$$

$$+ k_{r2}(\phi_s - \phi_{r2} - \phi_{NRS}) + c_{r2}(\dot{\phi}_s - \dot{\phi}_{r2})$$

for $i = 1, 2$ $\phi_{NRS} = \dfrac{\beta_{Li} L_{DLi} - \beta_{Ri} L_{DRi}}{St_{fi}}$

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5. Front axle → unsprung mass – DOF-5

$$M_{1f}\ddot{z}_{1f} + k_f(z_{1f} - (z_c - a\theta))$$

$$+ c_f(\dot{z}_{1f} - (\dot{z}_c - a\dot{\theta}))$$

$$+ k_{tf}(z_{1f} - q_f) = 0$$

6. Front axle roll

$$-\left(I_{xx}\ddot{\phi}_f + m_f h_2^2 \ddot{\phi}_f\right)$$

$$= -m_f g h_{us} (\phi_i) + k_{rf}(\phi_f + \phi_{road} - \phi_s)$$

$$+ c_{rf}(\dot{\phi}_f - \dot{\phi}_s) + k_{tri}(\phi_f - \phi_{road})$$

$$\phi_{road} = \frac{Z_f}{W T_f}.$$

6. Rear - 1 → DOF - 6

$$M_2 \ddot{Z}_2 - k_2 (Z_c - Z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR_2} + l_2 \theta)$$

$$- c_2 (\dot{Z}_c - \dot{Z}_2 + l_2 \dot{\theta})$$

$$+ k_{t2} \left(Z_2 - \left(\frac{Z_{rL2} + Z_{rR2}}{2} \right) \right) = 0$$

7. Rear - 2 → DOF - 7

$$M_3 \ddot{Z}_3 - k_2 (Z_c - Z_3 - \beta_{L3} - L_{DL3} - \beta_{R3} L_{DR3} + l_3 \theta)$$

$$- c_2 (\dot{Z}_c - \dot{Z}_3 + l_3 \dot{\theta})$$

$$+ k_{t2} \left(Z_3 - \left(\frac{Z_{rL3} + Z_{rR3}}{2} \right) \right) = 0$$

So, β_{L2} β_{L3} L_{DL2} L_{DR2}

β_{R2} β_{R3} L_{DR3} L_{DR3}

↓

4 eqn are solved to get these

4. Rear axle ~~pitch~~ . roll :- → r_2 → middle axle

$$- \left(I_{xx} \ddot{\phi}_2 + m_2 h_2^2 \ddot{\phi}_2 \right)$$

$$= - m_2 g h_{us} \left(\phi_2 + \phi_{NRS} \right)$$

$$+ k_{r2} \left(\phi_2 + \phi_{NRS} - \phi_s \right)$$

$$+ c_{r2} \left(\dot{\phi}_2 - \dot{\phi}_s \right)$$

$$+ k_{tri} \left(\phi_2 + \phi_{NRS} - \phi_{road} \right)$$

$$\phi_{road} = \frac{Z_{rLi} - Z_{rRi}}{WT_i}$$

5. Rear axle roll → r_3 → rear most

$$-\left(I_{xx}\ddot{\phi}_2 + m_2 h_2^2 \ddot{\phi}_2\right)$$

$$= -m_2 g h_{us}\left(\phi_2 + \phi_{NRS}\right)$$

$$+ k_{r3}\left(\phi_3 + \phi_{NRS} - \phi_s\right) + C_{r3}\left(\dot{\phi}_3 - \dot{\phi}_s\right)$$

$$+ \cancel{C}\, k_{ti}\left(\phi_3 + \phi_{NRS} - \phi_{road}\right)$$

camlin

Date :

shock absorber :

$F_S = -c\dot{r} + k(r-L) + F_{preload}$ $r \neq L$

$F_S =$ preload.

2P.